



KLE Technological
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Department
of
Electronics and Communication Engineering

Mini Project Report
on
**MOBILE CHARGER WITH MULTIPLE VOLTAGE
REGULATION & PROTECTION TECHNIQUES**

By :

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING

CERTIFICATE

This is to certify that project entitled “Mobile Charger with Multiple Voltage Regulation and Protection Techniques” is a Bonafide work carried out by the student team “ (Suraj – 01FE23BEC287, Sahana-01FE23BEC286,Pavitra–01FE23BEC277, Sagar – 01FE23BEC251)”. The project report has been approved as it satisfies the requirements with respect to the Mini project work prescribed by the university curriculum for BE (V semester) in Department of Electronics and Communication Engineering of KLE Technological University for the academic year 2025- 26.

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ABSTRACT

The rapid growth in the use of mobile devices has increased the demand for reliable and efficient charging solutions. This project focuses on the design and implementation of a mobile charger that provides a regulated DC output suitable for charging mobile phones. The proposed system converts the available AC supply into a stable DC voltage using a step-down transformer, rectifier circuit, filtering components, and a voltage regulation stage. Protection mechanisms are incorporated to ensure safe operation during over-current and short-circuit conditions. The designed charger delivers a constant output voltage required for mobile charging applications. Hardware implementation and testing demonstrate that the circuit operates reliably and meets the desired specifications. This project provides a simple, low-cost, and effective solution for mobile charging and also serves as a practical learning platform for understanding power supply design concepts in electronics.

Keywords: Mobile charger, Regulated power supply, AC to DC converter, Rectifier circuit, Voltage regulator, Short protection circuit.

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Chapter 1

1. Introduction

A mobile charger is an electronic system that supplies power to mobile devices by converting the input electrical energy into a suitable and regulated DC form required for battery charging. The main purpose of a charger is to deliver a controlled and stable output voltage and current so that the connected device can be charged safely and efficiently. Proper voltage regulation and current limiting are important to protect the battery as well as the internal circuits of the mobile device from electrical stress and damage.

Mobile charging systems are generally constructed using basic power supply blocks such as a step-down transformer, rectifier, filter circuit, and voltage regulator. In this project, these stages are implemented using discrete electronic components. A bridge rectifier and filter capacitor are used to convert AC input into smooth DC output, while voltage regulation is achieved using linear regulators and a zener diode-based regulator circuit. The Zener diode provides a stable reference voltage and also plays an important role in controlling the operation of the protection circuits. The design includes multiple maximum current handling circuits to support different load requirements.

Along with power conversion, safety and reliability are key aspects of charger design. This project incorporates multiple short-circuit and over-current protection techniques using different approaches. A BJT-based protection circuit is used for higher current operation, while a MOSFET with optocoupler-based protection circuit is implemented for lower current applications. These protection circuits detect fault conditions and disconnect or limit the output current, thereby preventing damage to the charger and the connected device. The complete charger circuit is simulated, implemented on PCB, and tested successfully to verify its performance under normal and fault conditions.

1.1 Motivation

Modern mobile chargers available in the market are compact, efficient, and reliable due to the use of advanced integrated circuits. However, their internal design is highly complex and difficult to understand at a basic level. This creates a need to study the fundamental working principles of mobile charging systems through a simple and practical design that clearly demonstrates the essential power supply stages.

The motivation of this project is to design and implement a mobile charger using discrete electronic components such as transformers, rectifiers, filter circuits, voltage regulators, and Zener diode-based regulation. The project also focuses on understanding safety aspects by implementing multiple maximum current handling circuits and different short-circuit protection techniques using BJT and MOSFET-based circuits. Developing the charger on PCB provides practical exposure to power supply design, testing, and protection concepts, forming a strong foundation for advanced electronic system design.

1.2 Objectives

- i. To design and simulate a mobile charger circuit that provides a regulated 5 V DC output suitable for mobile charging applications.
- ii. To convert the available AC input supply into DC voltage using a step-down transformer, bridge rectifier, and filter capacitor.
- iii. To implement voltage regulation using linear voltage regulators and a Zener diode-based regulation circuit to maintain a stable output voltage.
- iv. To incorporate multiple short-circuit and over-current protection techniques using BJT-based and MOSFET with optocoupler-based protection circuits.
- v. To fabricate the designed charger circuit on a printed circuit board (PCB) with proper component selection and layout considerations.
- vi. To experimentally test and verify the performance of the charger under normal operating conditions and during fault conditions such as short circuits.

1.3 Problem Statement

To design, simulate, and implement an AC–DC mobile charging circuit that converts a 230 V, 50 Hz supply into a stable 5 V DC output using discrete electronic components. The system uses a transformer, bridge rectifier, filter circuit, and Zener-based voltage regulation to obtain low-ripple DC output. The project incorporates multiple current handling circuits and BJT- and MOSFET-based short-circuit protection mechanisms to ensure safe and reliable operation. The complete circuit is implemented on PCB and tested to verify stable output and effective protection.

1.4 Organization of the Report

Chapter 1 : Introduces the project and presents the motivation, objectives, and problem statement related to the design of a mobile charger with protection circuits.

Chapter 2 : Discusses the electronic system design, including alternative solutions and the functional block diagram of the proposed charger.

Chapter 3 : Describes the implementation details, specifications, and system design methodology followed for simulation and PCB realization.

Chapter 4 : It presents the simulation analysis and hardware results obtained from the implemented circuit

Chapter 5 : Concludes the report with a summary of the work carried out and discusses the future scope and societal applications of the project.

Chapter 2

2. Electronic System design

2.1 Alternative Solutions

1. SMPS – Switched Mode Power Supply

Most of the advanced mobile chargers available today use a technique known as a Switched Mode Power Supply (SMPS). In an SMPS-based charger, the input AC voltage is first converted into DC and then switched at high frequency using power electronic switches such as MOSFETs. This high-frequency switching allows the use of smaller transformers and passive components, resulting in compact size and high efficiency. SMPS chargers offer several advantages such as reduced power loss, lighter weight, and improved efficiency compared to linear power supply-based chargers. They are capable of delivering higher output power while generating less heat, making them suitable for fast charging and portable applications. Due to these benefits, SMPS technology is widely used in commercial mobile chargers.

However, SMPS circuits are complex in design and operation. They require precise control circuitry, high-frequency switching techniques, electromagnetic interference (EMI) considerations, and careful PCB layout. This complexity makes SMPS-based designs difficult to analyze and understand at a fundamental level. For this reason, although SMPS is an effective solution, it is not chosen in this project, where the focus is on understanding basic power supply operation, voltage regulation, and protection mechanisms using discrete components.

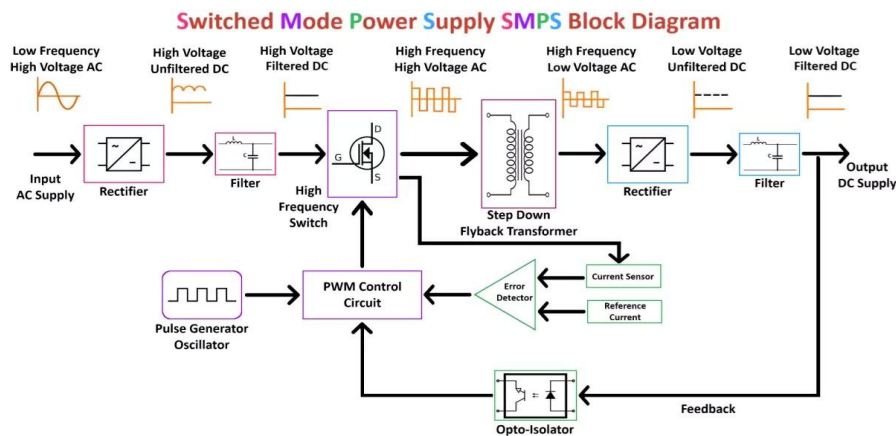


Figure 1 : Reference SMPS block diagram (source : hackatronic.com)

2. Linear power supply circuit

A linear power supply circuit is commonly used to provide a stable and regulated DC output for low-power electronic applications. In this method, the AC mains supply is first stepped down to a lower voltage using a transformer and then converted into DC using a bridge rectifier. A filter capacitor is used to reduce ripple and smooth the rectified output, producing a near-DC voltage suitable for further regulation.

The regulated output is obtained using linear voltage regulators or zener diode-based regulation circuits, which maintain a constant output voltage under varying load conditions. Linear power supply circuits are simple in design and easy to analyze, allowing protection features such as current limiting and short-circuit protection to be implemented using discrete components. Although their efficiency is lower compared to SMPS, linear power supplies offer stable output and low noise, making them suitable for educational and experimental charger designs like the one implemented in this project.

2.2 Functional block diagram

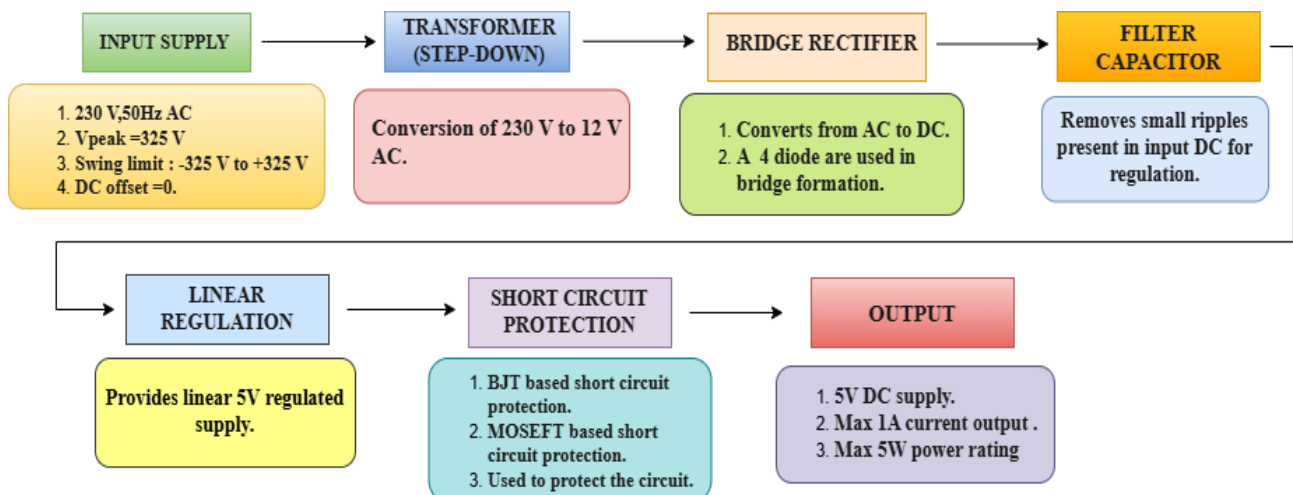


Figure 2 : Functional Block diagram

Chapter 3

3. Implementation details

3.1 Specifications

1. Input Supply

- Single Phase 230V, 50Hz AC input
- Ac input swings between “ -325 V to +325 V”

2. Output

- 5V $\pm 5\%$ Regulated DC power supply.
- Current rating or Maximum current is 1A.
- Output voltage ripple < 50mV

3. Transformer Specifications

- 12 V – 0 V – 12 V center tapped transformer.
- Maximum current rating = 1A.

4. Regulator Specifications

- 5V output supply.
- 1A maximum current.

3.2 System Design

The system design of the proposed mobile charger is based on a linear regulated power supply architecture. The complete system is divided into functional blocks, each performing a specific role in converting the AC mains supply into a stable and protected 5 V DC output. The major blocks include the step-down transformer, bridge rectifier, filter capacitor, voltage regulation circuit, and short-circuit protection circuit. Each block is designed and selected based on the required electrical specifications and safety considerations.

3.2.1 Transformer

The transformer is the first block in the system design and is used to step down the input 230 V, 50 Hz AC mains supply to a lower AC voltage suitable for the charger circuit. In this project, a centre-tapped 12 V – 0 – 12 V transformer is used. It provides electrical isolation between the mains and the low-voltage circuitry, ensuring safe operation.

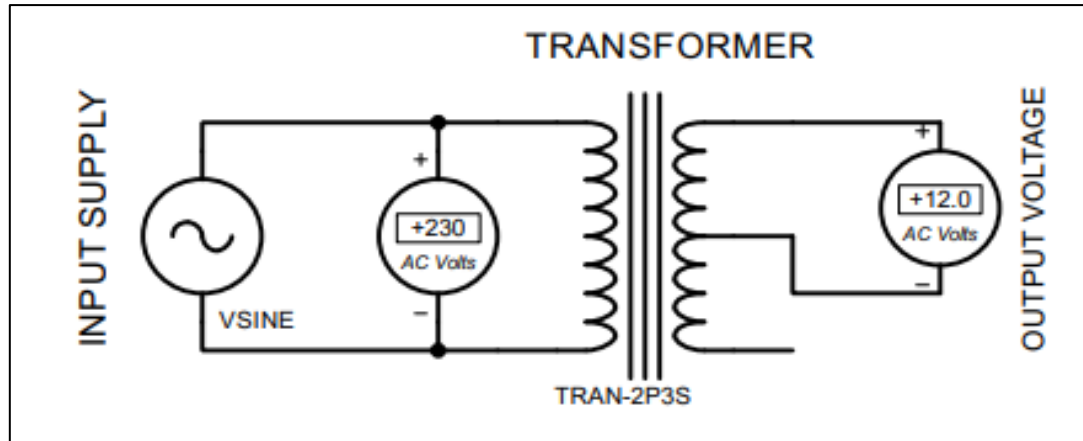


Figure 3 : Transformer

The transformer works on the principle of mutual induction, where an alternating magnetic field produced in the primary winding induces an AC voltage in the secondary winding. The voltage relationship is given by $\frac{V_p}{V_s} = \frac{N_p}{N_s}$, and for this design, the turns ratio is approximately 19:1. The transformer has a current rating of 1 A and a power rating of 12 VA, which is sufficient to meet the load requirements. The 12 V AC output obtained is supplied to the bridge rectifier for further processing.

3.2.2 Bridge Rectifier

Bridge rectifier is used to convert 12 V AC voltage from Transformer secondary output to pulsating DC voltage. The bridge rectifier is preferred over other rectifier circuits because it utilizes both half cycles of the AC input, resulting in higher efficiency, reduced ripple content, and simple two-wire input operation. The bridge rectifier is employed using diodes 1N4007 for the circuit of 7805 regulator and Zener diode regulator as these circuit need 1A max current. The diodes 1N5408 for the circuit based on regulator LM323. In bridge Rectifier there are 4 diodes placed as shown in Fig 3.2.2 and for every cycle two diodes will be ON , providing output similar to the full wave rectifier. Here the output supply will be having 100 Hz as frequency.

The forward voltage drop is the voltage required to start the conduction. According to datasheets the diode 1N4007 typically has the forward voltage drop of 0.7V at normal operation, but at the maximum current of 1A the voltage drop may go till 1V also. According to this whenever the bridge rectifier conducts it will turns on two diodes and the total voltage drop across input to bridge rectifier and output of bridge rectifier is additive voltage drop of both diodes. The input for bridge rectifier is the V_{peak} voltage and it is given by,

$$V_{peak} = V_{RMS} \times \sqrt{2}$$

As $V_{RMS} = 12 \text{ V}$, the V_{peak} will be 16.97V

Therefore the bridge rectifier can provide the maximum output of

$$V_{OUT} = V_{peak} - (2 \times 0.7) = 15.57 \text{ V}$$

While selecting the diode , as the diode sees the reverse voltage when AC reverses and it is called as peak inverse voltage (PIV) given by $PIV \approx 2 \times V_{peak}$. Here peak voltage is around 16.97 V and there for the PIV of diode should be greater than 35 V. The forward current is based on the regulator and circuit requirement, therefore for 1A current diode 1N4007 and for 3A current diode 1N5408 is used. Both of these also have PIV of 1000 V and forward voltage drop (F_V) of 0.7 V. The output maximum voltage would be 15.57 V.

3.2.3 Filter capacitor

The output obtained from the bridge rectifier is a pulsating DC voltage containing ripple. To reduce these ripples and obtain a smooth DC voltage, a filter capacitor is connected at the rectifier output. The capacitor charges to the peak value of the rectified voltage and discharges between cycles, thereby reducing voltage fluctuations and improving the quality of the DC supply.

The value of the filter capacitor depends on the load current, ripple frequency, and allowable ripple voltage and is calculated using $C = \frac{I}{f\Delta V}$. Since the ripple frequency of the bridge rectifier is 100 Hz and the maximum load current is 1 A, an allowable ripple voltage of 2 V is selected to ensure proper operation of the 7805 regulator. Based on this, a capacitor value of approximately 5000 μF is obtained, and a standard 4700 μF electrolytic capacitor is used. This capacitor ensures stable input to the regulator and improves overall performance of the charger circuit.

i.e. for 1A, capacitance is given by ,

$$C = \frac{(I = 1)}{(f = 100) \times (\Delta V = 2)} = 5000\mu F \approx 4700\mu F$$

The maximum load current for the LM323 regulator is 3 A, a slightly higher allowable ripple voltage is selected to balance capacitor size and power dissipation. Based on the design requirement, a capacitor value of approximately 3700 μF is chosen. Accordingly, a 3700 μF electrolytic capacitor is used at the input of the LM323 regulator to provide adequate ripple reduction and ensure stable operation under high current conditions. This capacitor helps maintain a reliable DC input to the regulator while supporting the higher current demand of the 3 A output circuit.

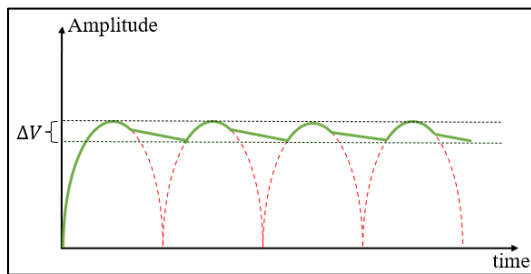


Figure 4 : with filter capacitor

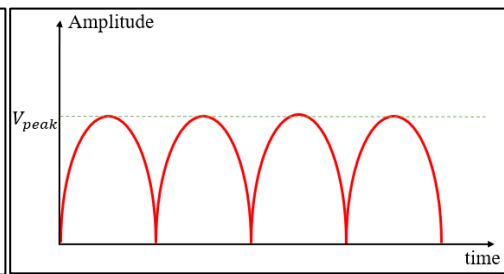


Figure 5 : without filter capacitor

3.2.4 Regulator

3.2.4.1 Regulator using Zener Diode

The 1N4734A is a 5.6 V Zener diode used for voltage regulation by operating in its reverse breakdown region. When the input voltage applied across the Zener diode exceeds its breakdown voltage, the diode maintains an approximately constant voltage across the load. A series resistor is connected in series with the Zener diode to limit the current flowing through it and to protect the diode from excessive power dissipation. The Zener regulator provides a simple and low-cost method of voltage regulation and is commonly used as a reference voltage source in low-power circuits.

To improve the current handling capability of the Zener regulator, a BD139 NPN transistor is used as a series pass transistor. In this configuration, the Zener diode provides a stable reference voltage to the base of the transistor, while the transistor supplies the required load current. This reduces the current stress on the Zener diode and allows the circuit to drive higher loads compared to a Zener-only regulator. When the Zener diode enters breakdown at 5.6 V, it fixes the base voltage of the BD139 transistor. The transistor then operates in the active region and regulates the output voltage. The regulated output voltage is given by

$$V_{out} = V_Z - V_{BE}$$

where V_Z is the Zener breakdown voltage and V_{BE} is the base-emitter voltage of the transistor (approximately 0.7 V). Therefore, the output voltage is approximately 4.9 V to 5V, which is close to the required 5 V output.

Despite its simplicity, the Zener diode-based regulator has several drawbacks. The output voltage is highly dependent on load current and input voltage variations, resulting in poor line regulation and load regulation. Significant power is dissipated in the series resistor and the pass transistor, reducing overall efficiency. Additionally, Zener regulators have limited current handling capability, poor thermal stability, and higher output ripple compared to integrated regulators. As the load current increases, the output voltage may drop significantly, making the circuit unsuitable for applications requiring stable voltage under varying load conditions.

Due to these limitations, dedicated linear regulators such as the 7805 and LM323 are used instead of relying solely on the Zener regulator. The 7805 provides a precisely regulated 5 V output with built-in features such as current limiting, thermal shutdown, and short-circuit protection, making it more reliable for 1 A applications. Similarly, the LM323 is used for high-current applications up to 3 A, offering better regulation, higher current capability, and improved thermal performance.

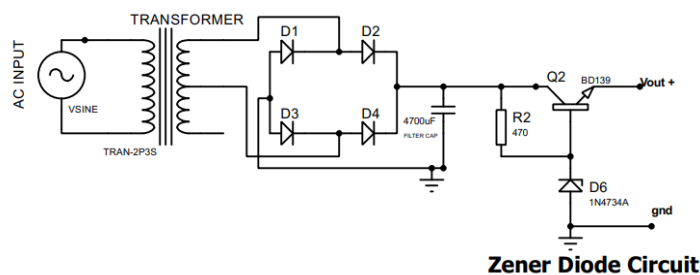


Figure 6 : Zener diode circuit

3.2.4.2 Regulator – LM7805

The 7805 is a fixed positive linear voltage regulator widely used to provide a stable 5 V DC output, making it well suited for mobile charging applications. In this project, the 7805 is used to regulate the filtered DC voltage obtained from the rectifier stage and deliver a constant output voltage regardless of variations in input voltage or load current. The regulator operates with an input voltage range of approximately 7 V to 25 V and is capable of supplying an output current of up to 1 A, which satisfies the current requirement of low-power mobile devices.

The operation of the 7805 is based on an internal reference voltage, error amplifier, and series pass transistor, which continuously adjust the conduction of the pass element to maintain a constant output voltage. The regulated output voltage is given by

$$V_{out} = 5 \text{ V}$$

as long as the input voltage remains above the minimum dropout requirement. To ensure stable operation, small capacitors are connected at both the input and output terminals of the regulator. The input capacitor helps suppress residual ripple from the filter stage, while the output capacitor improves transient response, reduces noise, and maintains output stability during sudden load changes.

The 7805 also includes built-in protection features such as internal current limiting, thermal shutdown, and short-circuit protection, which enhance the reliability and safety of the power supply. During excessive current or overheating conditions, the regulator automatically limits or shuts down the output to prevent damage. According to the datasheet, the maximum allowable junction temperature is 150°C, and a heat sink is provided to safely dissipate heat during higher load currents. Due to its simplicity, robustness, low noise, and built-in protection mechanisms, the 7805 is preferred in this project as a reliable and effective voltage regulator for delivering a stable 5 V DC output.

3.2.4.3 Regulator LM323

The LM323A is a fixed positive 5 V linear voltage regulator designed to supply high output current, making it suitable for applications requiring greater load capacity and improved current handling. In this project, the LM323A is used to provide a regulated 5 V DC output for high-current sections of the charger and associated protection circuits. The regulator operates with an input voltage range typically between 7.5 V and 15 V and is capable of delivering up to 3 A of output current when adequate heat sinking is provided.

The LM323A regulates the output voltage using an internal reference, control circuitry, and a high-current series pass transistor. It continuously adjusts the conduction of the pass element to maintain a constant output voltage even when the input voltage or load current varies. The regulated output voltage is given by

$$V_{out} = 5 \text{ V}$$

as long as the input voltage remains above the minimum required level. Due to its high current capability, the LM323A is particularly suitable for circuits where stable voltage must be maintained under heavy load conditions.

Since the LM323A is a linear regulator, excess voltage is dissipated as heat, and power dissipation increases with higher input voltage and output current. Therefore, the use of a proper heat sink is essential for safe and continuous operation. According to the datasheet, the maximum allowable junction temperature is 125°C, and effective thermal management is required to avoid overheating and thermal shutdown. The LM323A also incorporates internal protection features such as current limiting and thermal protection, enhancing system reliability. Owing to its robust design, high current capability, and stable regulation characteristics, the LM323A is preferred in this project for high-current 5 V output applications.

3.2.5 Output Capacitor

The output capacitor is connected across the regulated output of the voltage regulator to improve voltage stability and ensure smooth DC output. In this project, a 10 μ F capacitor is used at the regulator output to reduce residual ripple and suppress high-frequency noise that may be present due to rectification and regulation processes. This capacitor acts as a local energy storage element and supplies instantaneous current to the load during sudden load changes, thereby preventing momentary drops in output voltage.

The output capacitor also plays a crucial role in enhancing the transient response of the regulator. During rapid changes in load current, the regulator may not respond instantly; the capacitor compensates for this delay by providing or absorbing current as required. This helps in maintaining a stable output voltage and prevents oscillations or fluctuations at the regulator output. Proper selection of the 10 μ F output capacitor ensures reliable regulation, improved noise performance, and consistent 5 V DC output, making it suitable for mobile charging and low-power electronic applications.

3.2.6 Short Circuit Protection

Short circuit protection is an essential safety feature in power supply and mobile charging circuits to prevent damage caused by excessive current flow during fault conditions. A short circuit can result in sudden high current, leading to overheating and failure of components such as transformers, rectifiers, voltage regulators, and pass devices. To ensure safe and reliable operation, the proposed system incorporates dedicated short circuit protection circuits that continuously monitor the output current and disconnect or limit the supply when abnormal conditions are detected. In this project, multiple short circuit protection techniques are implemented to support different current levels, thereby improving system reliability and protecting both the charger and the connected load.

3.2.6.1. BJT -based Short Circuit Protection

A BJT-based short circuit protection (SCP) circuit is implemented to protect the power supply and connected load from damage caused by accidental short circuits. During a short circuit, the output current increases sharply, which can overheat and damage components such as the voltage regulator, diodes, transformer, and load. The SCP circuit detects this abnormal condition and automatically disconnects the load to ensure safe operation.

Components Used

- Q2 (CK100 – PNP transistor): Acts as the main pass transistor supplying power to the load.
- Q1 (BC547 – NPN transistor): Acts as the sensing transistor to detect abnormal current conditions.
- Resistors: Used for biasing, current limiting, and control of the transistors.

Sizing of the Resistors

➤ R8 = 150 Ω

- LED forward voltage ≈ 2 V
- Desired LED current ≈ 10 – 15 mA

$$R_8 = \frac{5-2}{15m} = 200\Omega \approx 150\Omega$$

➤ Coupling resistors R6=330Ω

- For NPN transistor , $I_C = \beta \times I_B$
- I_C = collector current (100mA –from datasheet)
- $\beta_{forced} = 10$ (chosen value)

$$I_B = \frac{100m}{10} = 10mA$$

$$V_{R_6} = V_{supply} - V_{BE} = 5 - 0.7 = 4.3 \text{ V}$$

$$R_6 = \frac{4.3}{10m} = 430\Omega \approx 330\Omega$$

➤ R7=10kΩ

- According to the thumb rule the standard pull down resistor value is 10k.

➤ R9=330Ω

- LED forward voltage ≈ 2 V
- Desired LED current ≈ 10 – 15 mA

$$R_9 = \frac{5-2}{10m} = 300\Omega \approx 330\Omega$$

➤ **Coupling resistor R5=1KΩ**

$$VR_5 = V_{supply} - V_{BE} = 5 - 0.7 = 4.3 \text{ V}$$

- Desired $I_B = 3\text{-}5\text{mA}$

$$R_5 = \frac{4.3}{4\text{mA}} = 1.075\Omega \approx 1\text{K}\Omega$$

▪ **Working of BJT-Based Short Circuit Protection**

Normal Operation

During normal operation, the load resistance remains within a safe range, and the output current flows normally. The output voltage remains stable and no abnormal condition is detected. Under these conditions, the sensing transistor Q1 (BC547) remains OFF. This allows the pass transistor Q2 (BC558) to remain ON and supply power to the load. As a result, current flows normally, the green LED (D7) glows to indicate proper operation, and the fault LED (D8) remains OFF. The charger operates correctly without interruption.

Short-Circuit Condition

During a short circuit, the load resistance becomes nearly zero, causing a sudden rise in output current and a drop in output voltage. This abnormal condition is detected by the sensing transistor Q1 (BC547), which turns ON. When Q1 conducts, it removes the base drive of the pass transistor Q2 (BC558), forcing Q2 to turn OFF. As a result, the load is disconnected from the supply, and current flow is stopped to protect the circuit. During this condition, the green LED (D7) turns OFF, while the fault LED (D8) turns ON to indicate the presence of a short circuit. The circuit remains in a protected state until the fault is removed.

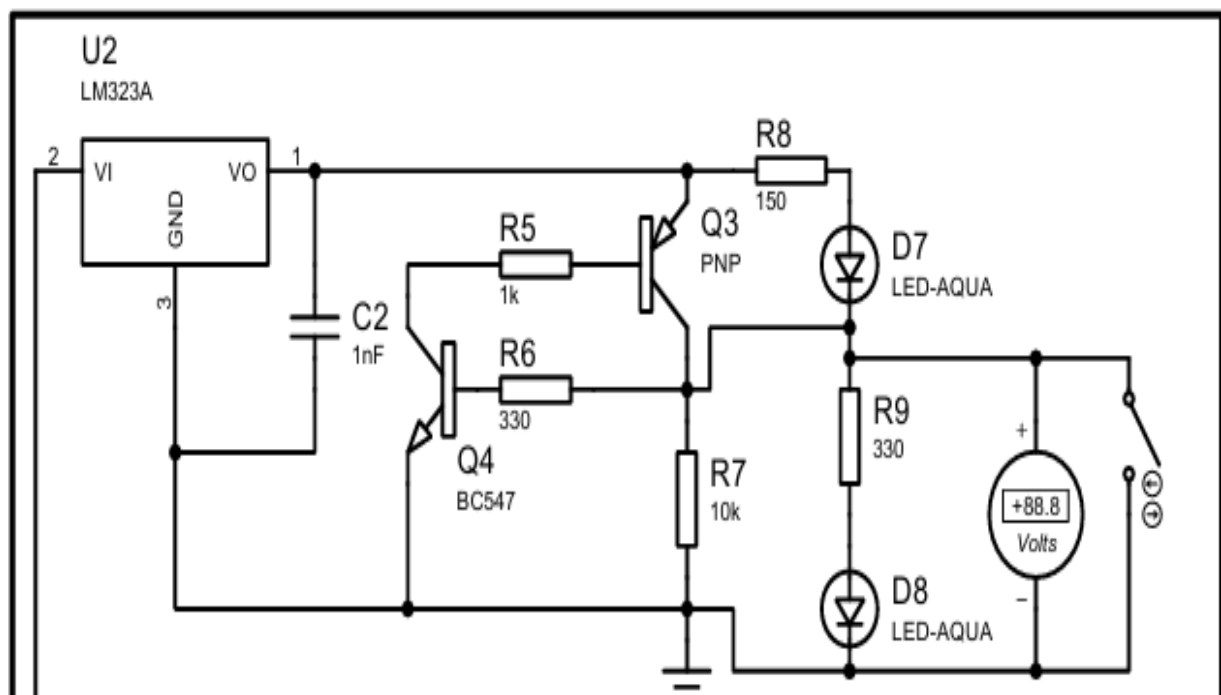


Figure 7 : BJT based short circuit protection

3.2.6.2 MOSFET-Based Short Circuit Protection

The MOSFET-based short circuit protection circuit is implemented in this project to prevent excessive current flow that may occur during overload or short-circuit conditions. Such abnormal conditions can cause damage to the power supply components and the connected load if not properly controlled. In this circuit, a power MOSFET is used as an electronic switching element to control the supply of current to the output. Under normal operating conditions, the MOSFET remains in the ON state and allows current to flow to the load.

An optocoupler is employed for fault detection and electrical isolation between the sensing circuit and the power path. When a short circuit or excessive current condition is detected, the optocoupler is activated, which in turn drives the MOSFET into the OFF state, disconnecting the load from the supply. The use of a MOSFET ensures fast response and low conduction losses, while the optocoupler provides isolation and improves system safety. This protection method enhances the reliability of the charger and effectively safeguards both the power supply and the connected devices.

Components Used

- IRFZ44N MOSFET – Used as a series pass switch to control power delivery to the load.
- PC817 Optocoupler – Used for fault sensing and electrical isolation.
- Resistors – Used for gate control, pull-down action, and current limiting
- LED (D5) – Used for short-circuit indication

Sizing of Resistors.

➤ **R1 =100K (Gate pull down resistors)**

According to the design rule, gate pull-down resistor = 47 kΩ to 220 kΩ. So 100k is standard and safe choice.

➤ **Resistor in series with LED D5**

- LED forward voltage $\approx 2\text{ V}$
- Desired LED current $=3\text{mA}$

$$R = \frac{5-2}{3m} = 1K\Omega$$

Working of MOSFET-Based Short Circuit Protection

During Normal Operation

During normal operation, the output voltage is healthy and the internal LED of the PC817 is not forward biased. As a result, the phototransistor remains OFF and does not affect the MOSFET gate. The MOSFET gate receives sufficient voltage from the output through the indicator LED and resistor network, resulting in $V_{GS} > V_{th}$. Hence, the MOSFET turns ON and behaves like a closed switch, allowing current to flow from the supply to the load. Since the phototransistor is OFF, the indication LED remains OFF, indicating normal operation.

Short-Circuit Condition

When a short circuit occurs, the output terminals are shorted and the load resistance becomes approximately zero. This causes the output voltage to collapse to nearly zero. Due to this voltage drop, the internal LED of the PC817 becomes forward biased and turns ON. The light emitted activates the internal phototransistor, which pulls the MOSFET gate towards ground. As a result, the gate-to-source voltage drops below the threshold voltage ($V_{GS} < V_{th}$), turning the MOSFET OFF. The MOSFET now behaves like an open switch, disconnecting the load from the supply and protecting the circuit from damage. At the same time, the indication LED turns ON, providing a visual indication of the short-circuit fault.

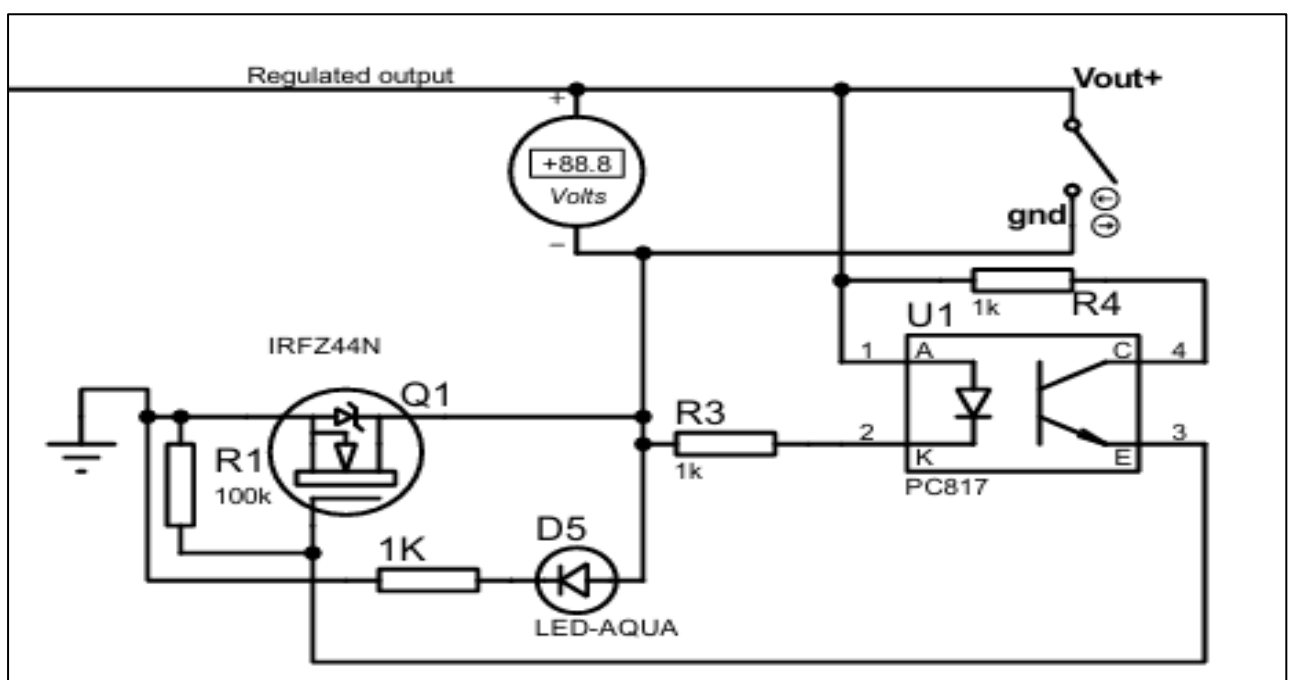


Figure 8 : MOSFET based short circuit protection

3.3 Flow Chart

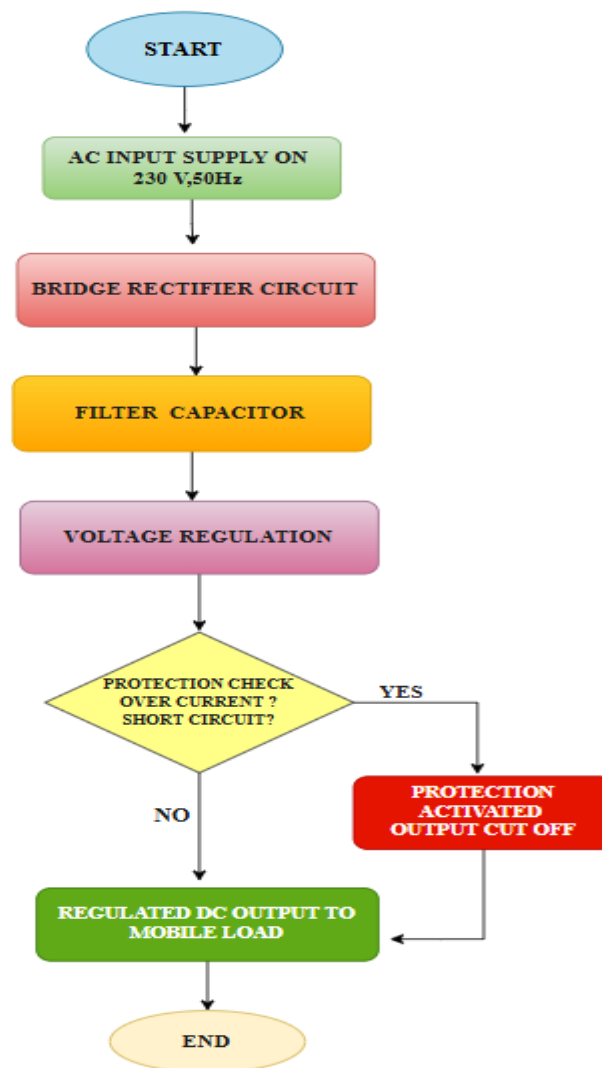


Figure 9 : Flow Chart

Chapter 4

4. Results and Discussions

This chapter presents the results obtained from both the simulation and hardware implementation of the proposed mobile charger circuit. The performance of each stage of the system—including the step-down transformer, bridge rectifier, filter capacitor, voltage regulators, and short-circuit protection circuits—is analyzed to verify correct operation. Simulation results are used to observe voltage levels, waveform behaviour, and regulator performance before hardware realization.

The hardware results are then discussed to validate the practical implementation of the circuit and to compare them with the simulated outcomes. Particular emphasis is placed on verifying proper AC to DC conversion, stable 5 V voltage regulation, and effective short-circuit protection under normal and fault conditions. The analysis confirms that the designed charger meets the specified requirements and operates reliably, demonstrating the correctness and effectiveness of the proposed system design.

4.1 Simulation Results

The complete mobile charger circuit was first designed and tested using simulation software to verify its correct operation before proceeding with hardware implementation. The simulation model includes all major functional blocks of the system, namely the step-down transformer, bridge rectifier, filter capacitor, voltage regulation circuits using the 7805, LM323, and Zener diode, as well as the short-circuit protection circuits. Simulating the entire system helped in understanding the behaviour of each stage and ensured that the design met the required specifications.

The simulation results show that the transformer successfully steps down the 230 V AC input to the required low-voltage AC level. The bridge rectifier converts this AC voltage into pulsating DC, and the filter capacitor effectively reduces ripple, providing a smooth DC input to the voltage regulators. The 7805 and LM323 regulators maintain a stable 5 V DC output under normal load conditions, while the Zener diode provides a reliable voltage reference for regulation and protection circuits. During simulated short-circuit conditions, the protection circuits operate as expected by limiting or disconnecting the output, thereby preventing excessive current flow. These results confirm the correctness of the design and provided confidence to proceed with hardware realization.

4.1.1 Bridge rectifier

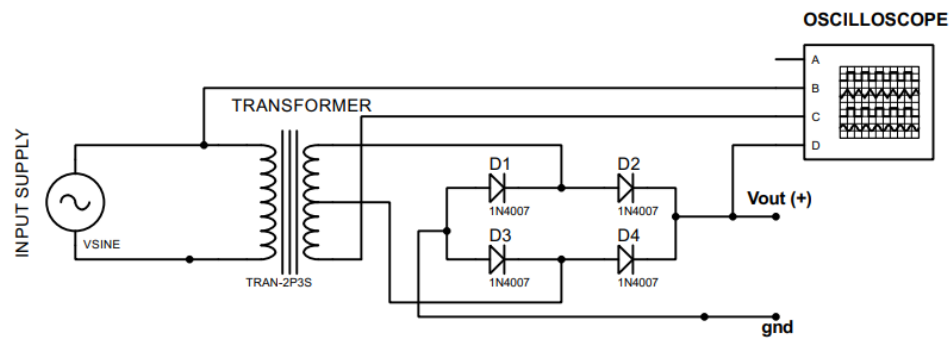


Figure 10 : Bridge rectifier

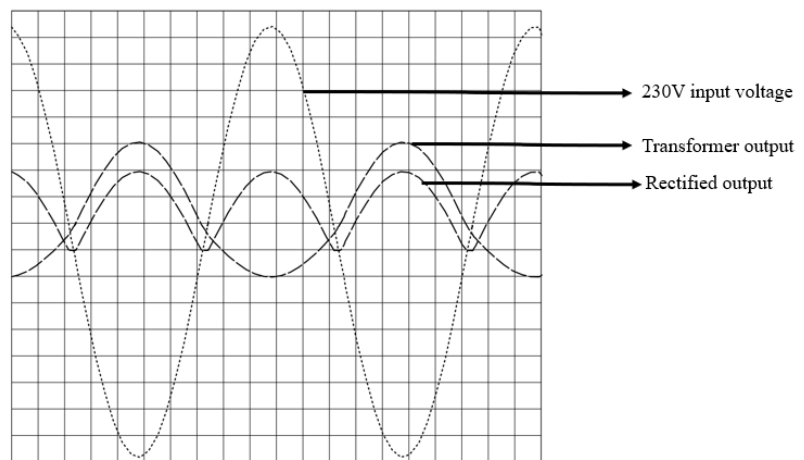


Figure 11 : output of bridge rectifier

4.1.2 Filter Capacitor

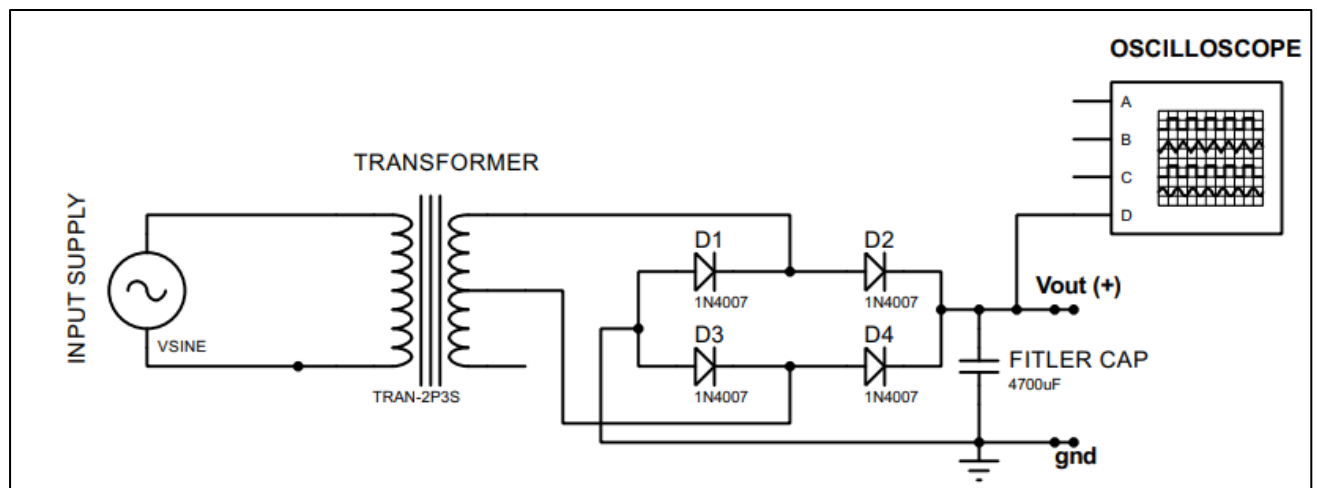


Figure 12 : Filter capacitor

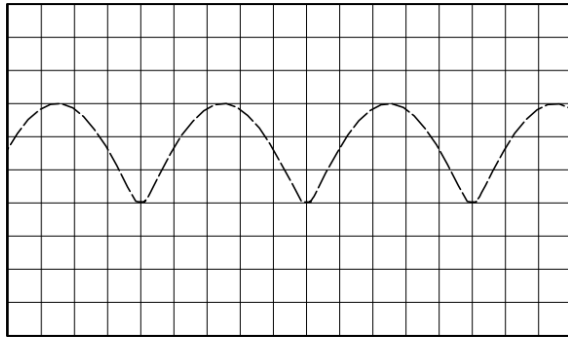


Figure 14 : without filter capacitor

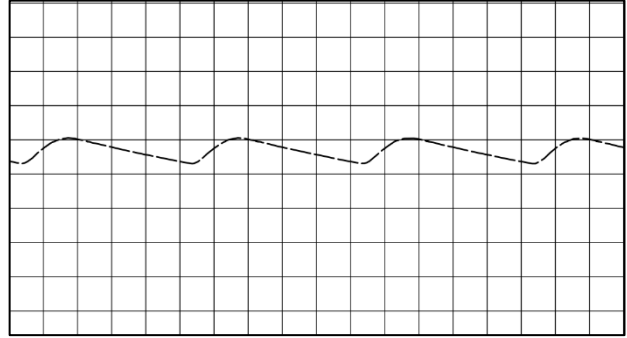


Figure 13 : with filter capacitor

Here in the images the voltage is of 5V per division for without filter capacitor and 2V per division for with filter capacitor. The simulated results clearly shows that the output voltage contains very small ripples.

4.1.3 Zener diode as regulator

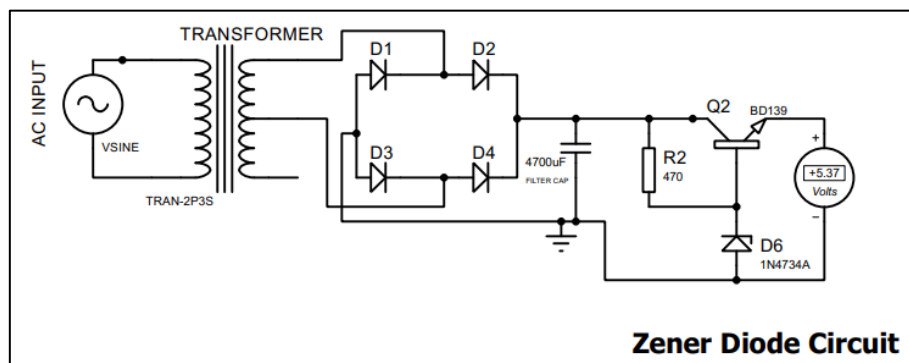


Figure 15 : Zener diode as regulator

4.1.4 LM7805 as voltage regulator

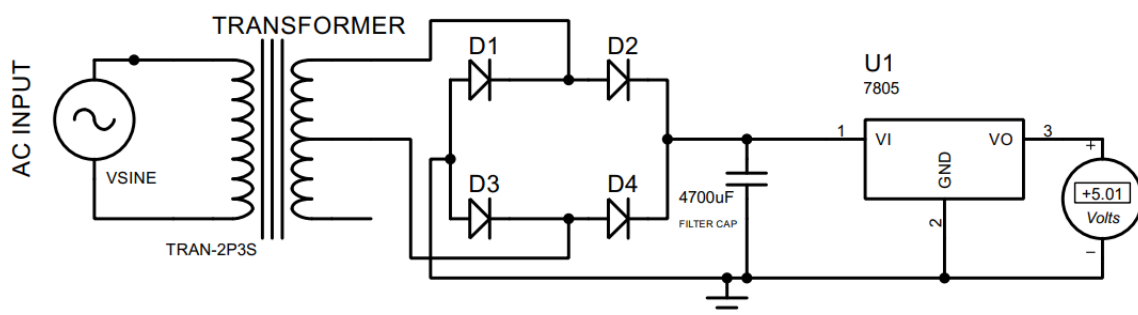


Figure 16 : LM7805 as regulator

4.1.5 LM323 as voltage regulator

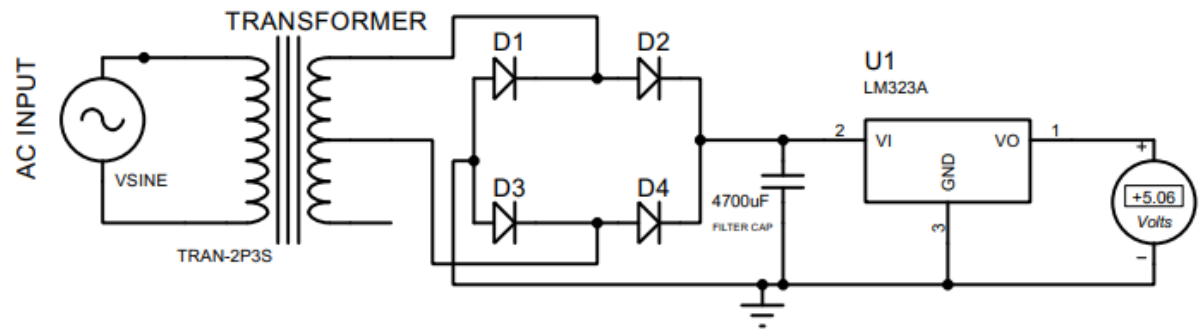


Figure 17 : LM323 as voltage regulator

4.1.6 Combined regulator circuit

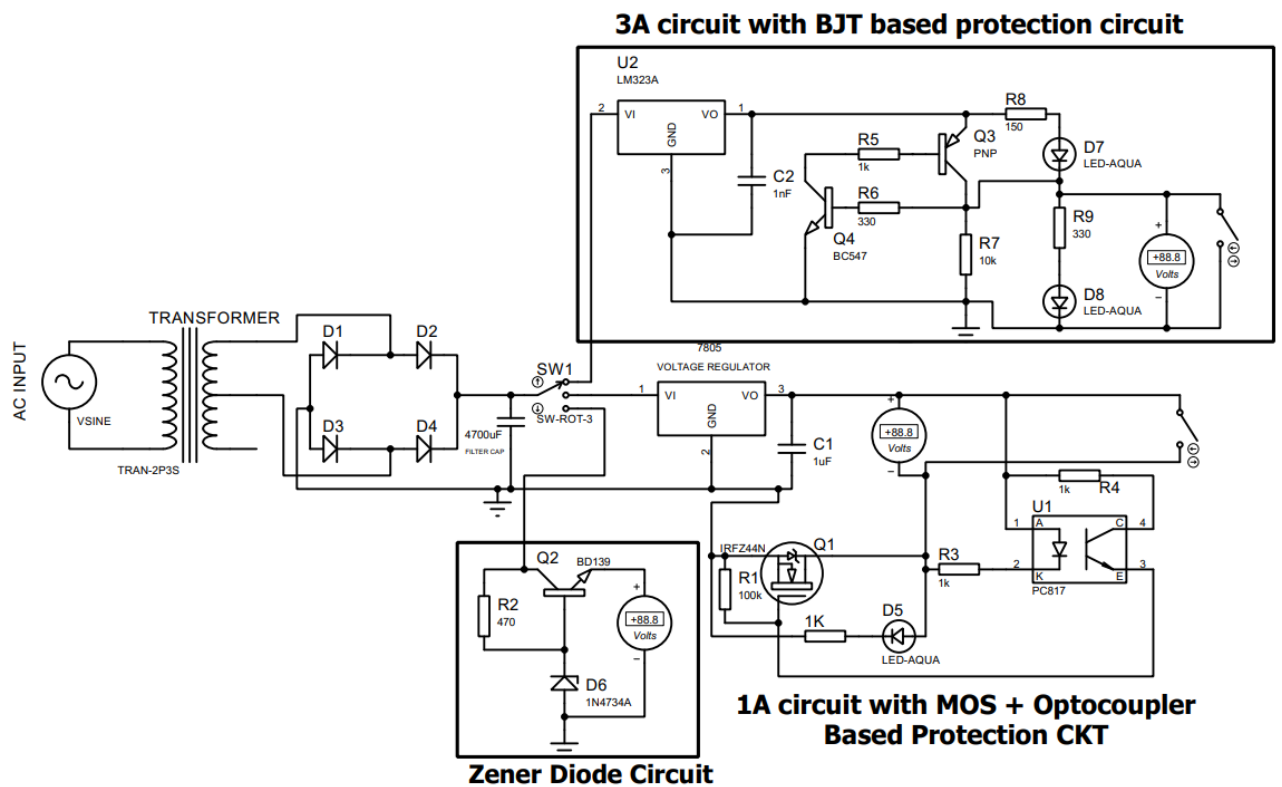


Figure 18 : combined circuit

4.2 Hardware realization

After successful simulation, the designed mobile charger circuit was implemented in hardware to validate its practical performance. The circuit was assembled on a general-purpose dotted PCB, where all components were mounted and interconnected according to the circuit diagram. Care was taken to ensure proper component orientation, secure soldering, and correct electrical connections. Heat sinks were provided for the linear regulators to manage power dissipation during operation.

The assembled circuit was powered using a 230 V AC input, and the output voltage and current were measured using standard measuring instruments. Under normal operating conditions, the circuit produced a stable 5 V DC output, confirming proper rectification, filtering, and voltage regulation. The short-circuit protection circuits were also tested by applying fault conditions, and the circuit successfully disconnected or limited the output, protecting the components from damage.

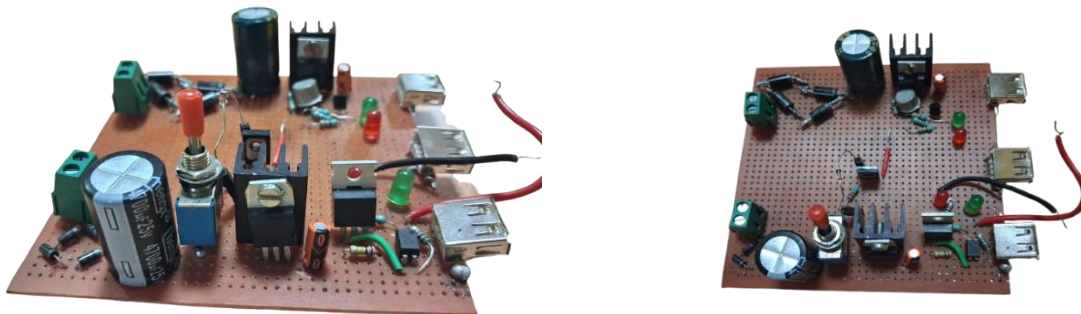


Figure 19 : Hardware Implementation

4.2.1 Zener diode PCB output

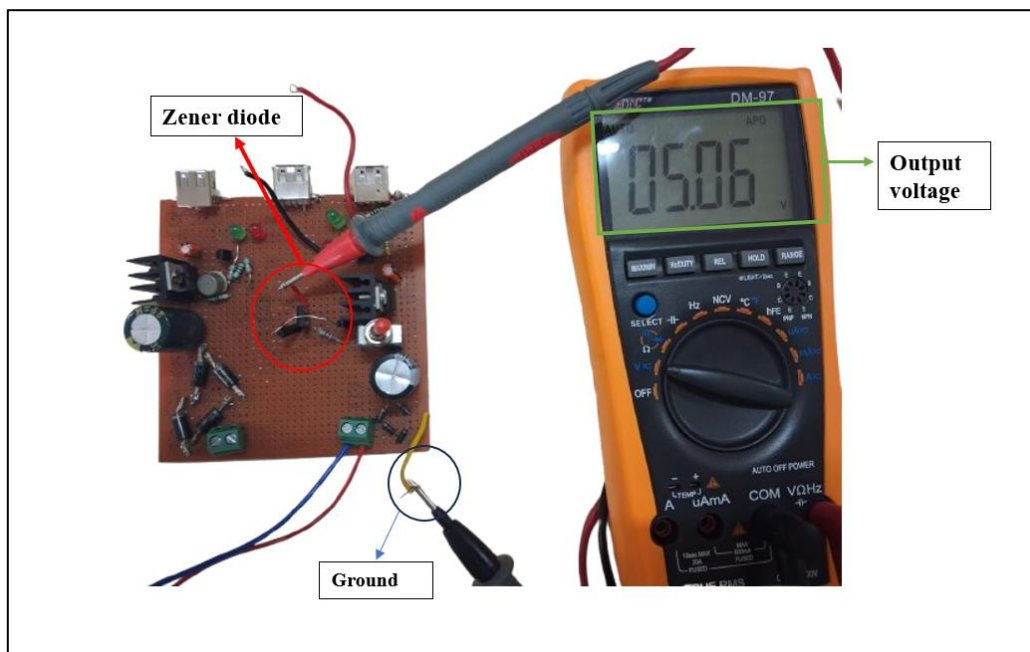


Figure 20 : Zener diode on hardware

4.2.2 LM7805 Regulator output

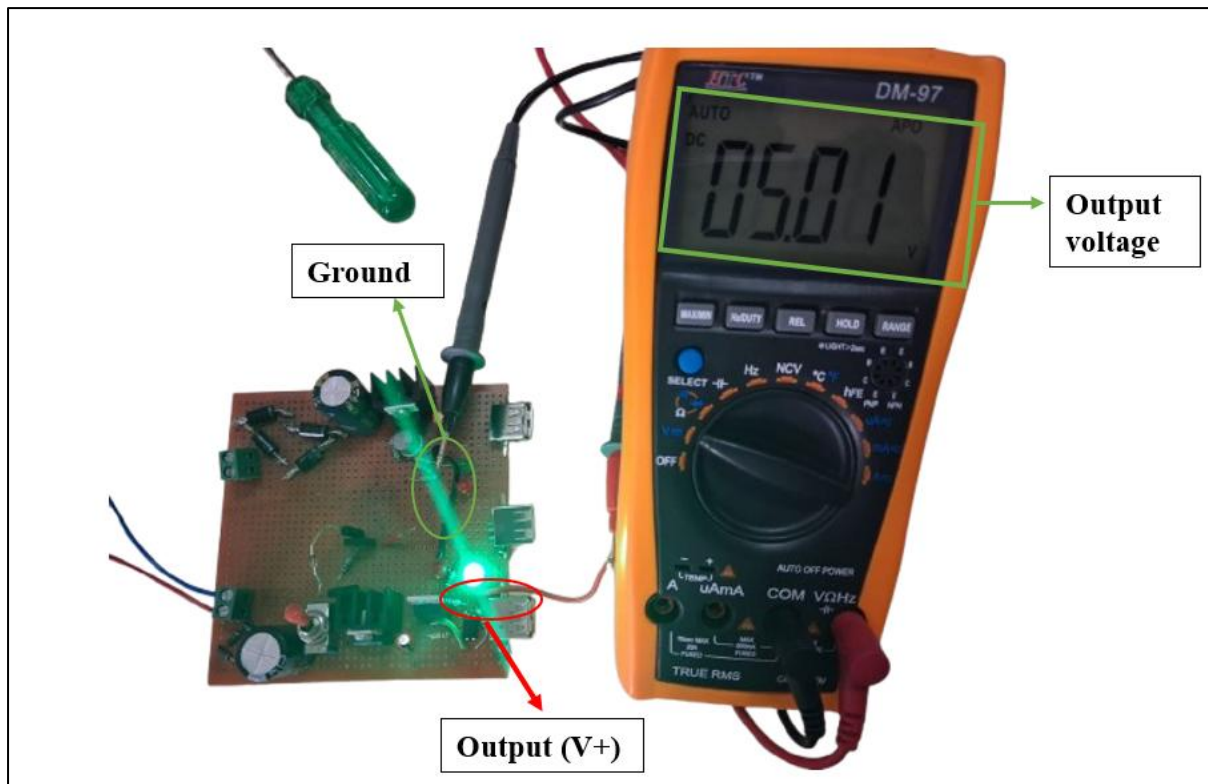


Figure 21 : LM7805 on hardware

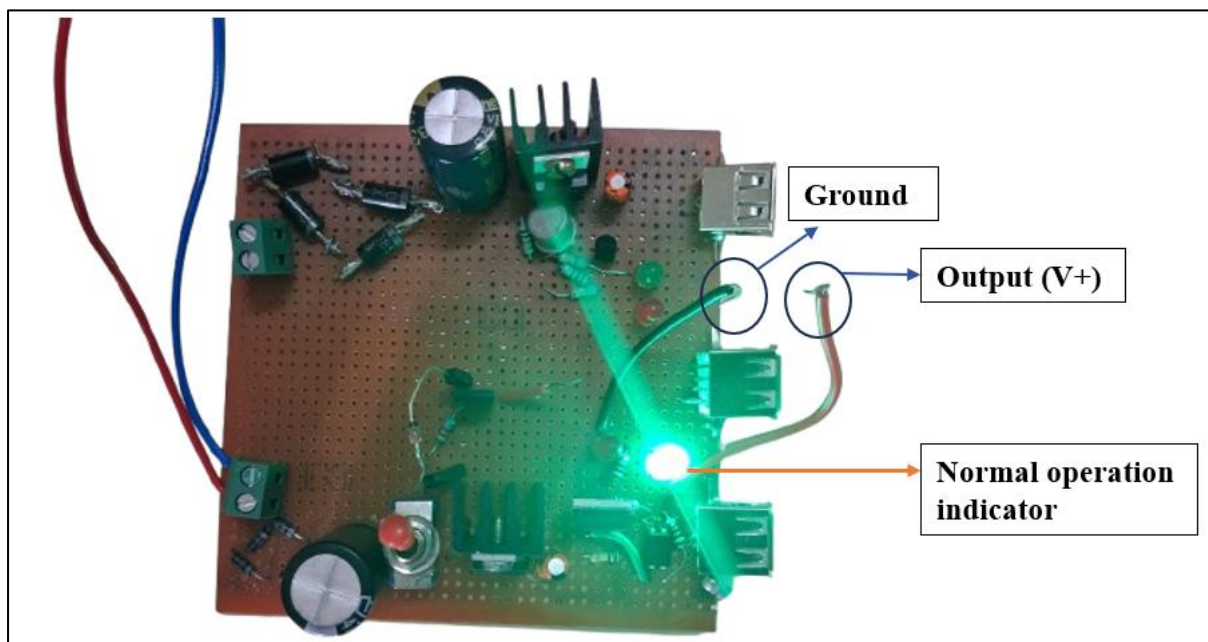


Figure 22 : Normal operation of the circuit

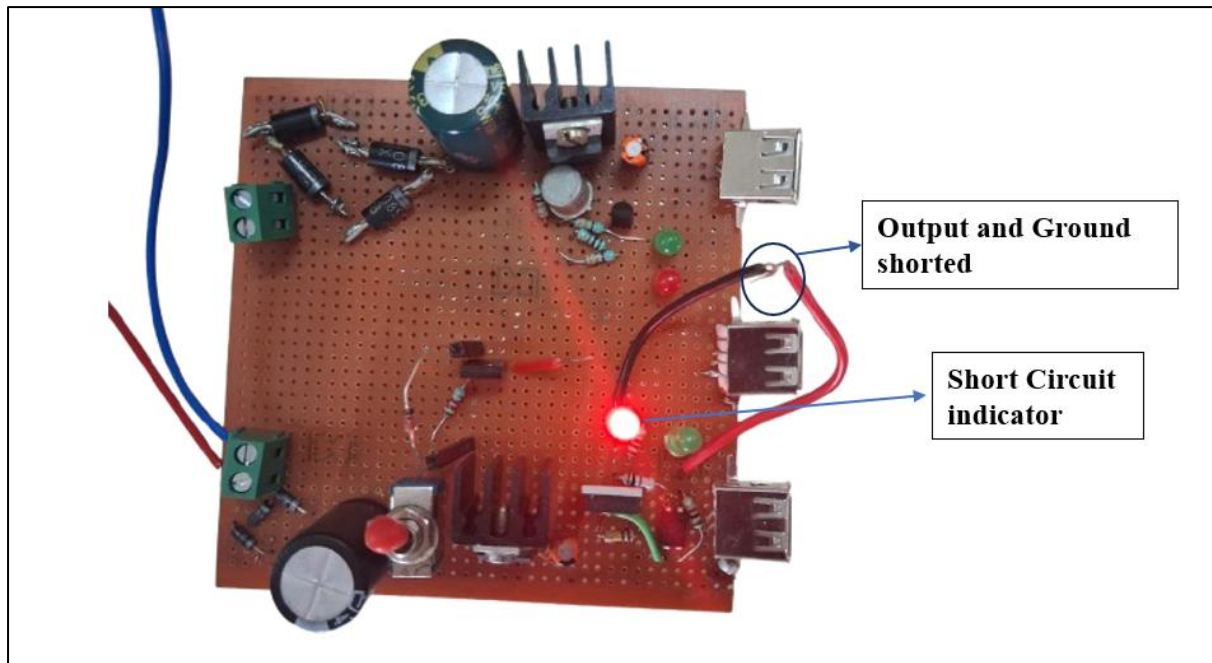


Figure 23 : Short Circuit condition

4.2.3 LM323 Regulator output

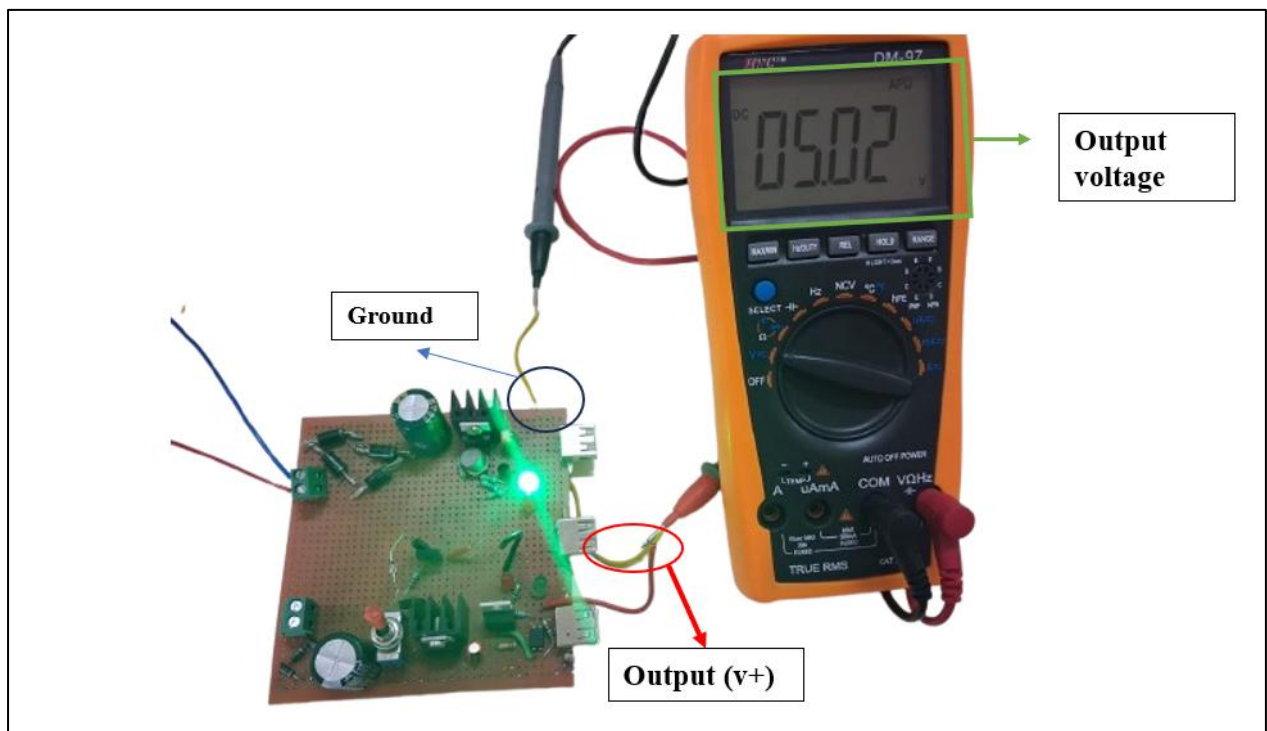


Figure 24 : LM323 on hardware

4.2.4 Charging a mobile phone



Figure 25 : mobile phone charging image

Chapter 5

5. Conclusion and reflections

5.1 Conclusion

In this project, a basic mobile charger system with effective short-circuit protection was successfully designed, simulated, and implemented. The charger converts 230 V AC mains supply into a regulated 5 V DC output suitable for mobile charging applications using a transformer, bridge rectifier, filter capacitor, and voltage regulation stages. Different regulation techniques, including IC-based regulators (7805, LM323A) and a Zener diode with BJT pass transistor, were studied and implemented to understand their performance, limitations, and practical applications. To enhance safety and reliability, both BJT-based and MOSFET-based short-circuit protection circuits were integrated and analyzed. The MOSFET-based protection, combined with an optocoupler, provided fast response, electrical isolation, and low power loss, making it more efficient and reliable compared to conventional protection methods. Visual indication using LEDs allowed easy identification of normal and fault conditions.

The hardware realization and testing confirmed that the charger delivers a stable output under normal operation and effectively disconnects the load during short-circuit conditions, thereby preventing damage to the power supply and connected devices. Overall, the project demonstrates a safe, reliable, and cost-effective mobile charger design and provides a strong foundation for further enhancement using advanced control techniques and smart monitoring features.

5.2 Future scope

Although the proposed mobile charger has been successfully implemented with over-current protection, short-circuit protection, and PCB realization, further enhancements can be made to improve performance and functionality. The system can be upgraded by replacing the linear regulation stage with a high-efficiency SMPS converter to reduce power loss and heat dissipation. Support for fast-charging standards and adaptive load detection can be incorporated to make the charger compatible with modern mobile devices.

Additional protection features such as over-voltage protection and thermal shutdown can be integrated to further enhance safety and reliability. The charger can also be extended with digital monitoring using a microcontroller to display output voltage, current, and fault status.

References

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